This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

FedMOP: Achieving Enhanced Privacy and Performance in Federated Learning via Momentum Orthogonal Projection

Yunlong Zhao[1] Xiaoheng Deng[1]* Hongyan Xu[1] Zhuohua Qiu[5] Xiaowen Hu[1]
Shan You[4] Yi Chen[2]* Chang Xu[4] Xiu Su[1]*
[1]Central South University [2]Hong Kong University of Science [3]SenseTime Research
[4]University of Sydney [5]Shenzhen University
{zhaoyl741, dxh, hongyanxu}@csu.edu.cn,{qiuzhuohua23}@email.szu.edu.cn, {vinaltria}@csu.edu.cn
{youshan}@sensetime.com, {yichen}@ust.hk,{c.xu}@sydney.edu.au, {xiusu1994}@csu.edu.cn

Abstract

Federated Learning (FL) faces a fundamental dilemma: existing defenses against gradient leakage attacks (GLAs) invariably sacrifice model performance for privacy protection through noise injection or gradient clip. We introduce Federated Learning with Momentum-Based Orthogonal Projection (FedMOP), a method that simultaneously achieves strong privacy guarantees and superior model performance. The key insight is to leverage initialization-based offset mechanisms that operate on orthogonal dimensions. For performance enhancement, FedMOP employs gradient orthogonal projection to counteract local drift, effectively offsetting each client's round-training initial model using global statistical context. For privacy protection, it introduces momentum-based trajectory offset hiding, which makes the offset vector inherently unrecoverable by constructing information barriers through private initialization and randomized evolution. These two mechanisms are synergistic rather than antagonistic. Theoretically, we prove convergence preservation and characterize the computationally infeasible inverse problem faced by attackers. Extensive experiments on CIFAR-10/100 and Tiny-ImageNet demonstrate that FedMOP not only defends effectively against state-of-the-art GLAs but also surpasses existing FL methods in both accuracy and convergence speed, validating its ability to jointly enhance privacy and performance in FL. Codes are available at https://github.com/zyl123456aB/FedMOP.

1. Introduction

Federated Learning (FL) [19] enables collaborative model training without exchanging raw data by sharing gradients across clients. Despite this privacy-preserving design, recent studies demonstrate adversaries can reconstruct sensitive training data by analyzing shared gradients [8, 17, 42], posing serious risks in privacy-sensitive applications such as medical diagnosis and financial analysis.

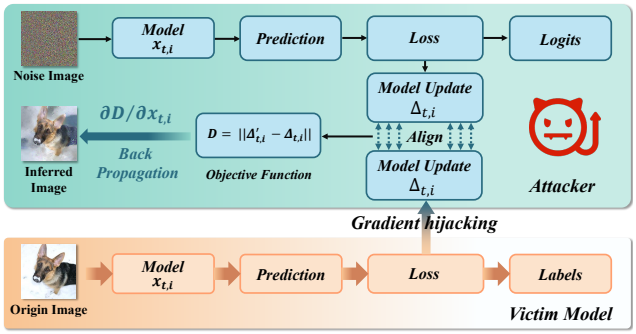


Figure 1. The framework of GLA. An attacker reconstructs private training data by minimizing the gradient difference between a dummy image and the victim model's true gradient.

This vulnerability has prompted extensive research into defense mechanisms, including differential privacy [1], gradient compression [27], and secure aggregation [3]. However, these approaches face a fundamental trade-off: stronger privacy guarantees often come at the expense of model performance due to noise injection, information loss, or computational overhead. Differential privacy requires substantial noise that significantly reduces accuracy on heterogeneous data; gradient compression loses critical components needed for convergence; secure aggregation incurs prohibitive cryptographic costs. Beyond privacy concerns, standard FL also suffers from slow convergence on non-Independent and Identically Distributed (IID) data due to local drift [12], where heterogeneous distributions cause divergence from the global optimum. Existing acceleration methods [14, 22] focus solely on performance without addressing privacy, creating a landscape where privacy and performance appear fundamentally antagonistic.

*Corresponding author is Xiaoheng Deng, Xiu Su, and Yi Chen

We bridge this gap by introducing Federated Learning with Momentum-Based Orthogonal Projection (FedMOP), which reconciles privacy and performance through a principled initialization-based approach. Our method builds on a key insight: by carefully controlling each client's *starting point* for local training, we can simultaneously correct for non-IID drift (performance) and obscure gradient information (privacy), without modifying the training process itself. Specifically, FedMOP shifts each client's initial model by a computed offset before local training begins.

For **performance**, the offset is designed via gradient orthogonal projection to be perpendicular to the local training direction. This ensures the shift incorporates global statistical information to counteract local drift, while orthogonality guarantees it does not interfere with gradient descent—enabling aggressive scaling for faster convergence.

For **privacy**, however, a naive implementation faces a critical vulnerability: if the offset is deterministically computable from observed models, an attacker can enumerate candidate values and solve gradient matching for each, breaking privacy through brute force. We address this by transforming the offset into a fundamentally uncomputable private trajectory. Each client initializes a secret random vector and evolves it through momentum dynamics, where each round's offset mixes the previous state with the current projection using stochastic coefficients. Reconstructing this trajectory requires inferring exponentially many unknowns—the initial secret vector plus all historical mixing parameters—creating a computationally intractable inverse problem.

Remarkably, momentum evolution strengthens both objectives: it provides variance reduction through exponential smoothing (improving convergence), while expanding the attacker's search space exponentially with training duration (strengthening privacy). This synergy enables FedMOP to achieve what prior methods could not: enhanced privacy *with* improved performance, rather than *despite* privacy protection. The key contributions of this work are as follows:

- We propose FedMOP with gradient orthogonal projection (performance) and momentum-based trajectory hiding (privacy) to achieve synergistic improvements.
- We introduce momentum-based hiding that transforms computable offsets into uncomputable trajectories, creating a high-dimensional inverse problem.
- We prove convergence preservation with $\mathcal{O}(1/(K^2T^2))$ rates, while establishing computational hardness bounds for offset reconstruction.
- Extensive experiments demonstrate 5-10× stronger privacy protection, 2-4% accuracy gains, and 1.5-2× faster convergence over existing defenses across multiple datasets and heterogeneity settings.

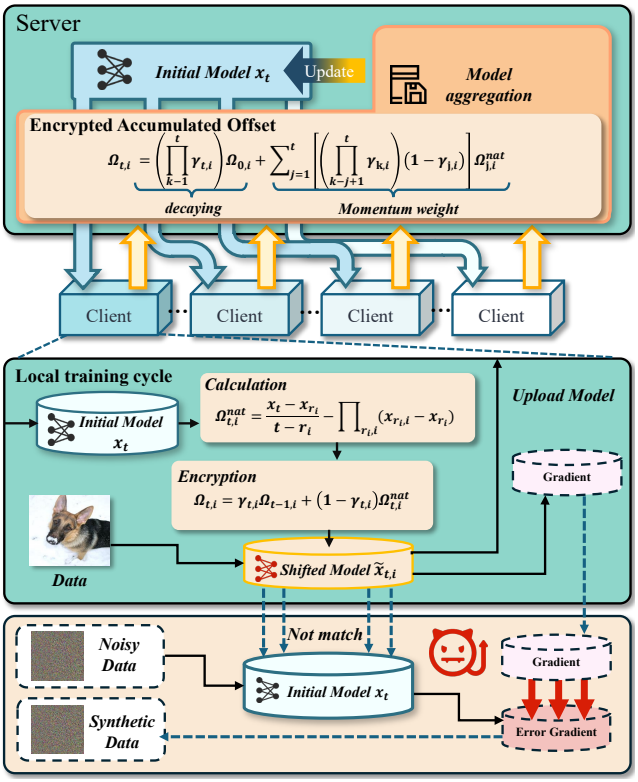


Figure 2. Framework overview. Unlike prior work where offsets are server-computable, our momentum-based trajectory hiding creates high-dimensional information barriers through private initialization and randomized evolution, while orthogonal projection ensures performance enhancement.

2. Related Work

Gradient Leakage Attack (GLA). GLA is a major privacy threat in FL, where shared gradients can be exploited to reconstruct private training data. Recent methods, such as Bayesian frameworks [2], gradient inversion techniques [7], and encryption-based defenses like the Key-Lock Module [23], have pushed forward gradient leakage mitigation. Traditional defenses include differential privacy [1], secure aggregation protocols [3], and gradient perturbation methods [10, 26]. Newer methods like federated distillation and homomorphic encryption are also gaining traction for preserving FL privacy.

Non-IID FL. Vanilla FL specifically FedAvg [19], underperforms in the presence of non-IID data [12, 15, 41]. Heterogeneous data distributions cause local updates to diverge from the global optimization objective, leading to local drift, which ultimately degrades global model performance and convergence speed. To mitigate these issues, various server-side corrections have been proposed, such as adjusting aggregation weights [31, 37] or enhancing global optimization using modified learning rates and momentum [11, 22]. Client-side approaches include adding globally relevant loss components [14] and utilizing virtual

data to reduce local biases and encourage knowledge sharing [28]. Additionally, recent studies have introduced adaptive weighting mechanisms that dynamically adjust each client's contribution, considering data diversity.

Gradient Orthogonal Projection (GOP). GOP has seen widespread use in multi-task and continual learning. In multi-task learning, it addresses conflicting gradients across tasks, as initially demonstrated in Gradient Surgery [38] and later refined by GradNorm [40], which dynamically balances task contributions. Orthogonal projections also enhance parameter sharing, ensuring efficient use of shared parameters across tasks [20]. In continual learning, Gradient Projection Memory (GPM) [24] helps retain crucial information from past tasks, thereby mitigating catastrophic forgetting. Subsequent methods [5, 35] refine projection spaces to minimize interference with prior task performance, allowing for more effective integration of new tasks.

3. Preliminary

As shown in Figure 1, while FL is designed to protect privacy by allowing collaborative training without sharing raw data, the shared gradients can still reveal detailed local data information, posing significant privacy risks. Consider the gradient update shared by the i-th client at round t. Let $g_{t,i} = \nabla F_i(x_t)$ represent the gradient of the local objective F_i with respect to the initial model x_t provided by the server. As shown in Figure 1, the attacker, typically a malicious server, initializes a synthetic input vector $z_{t,i}^{\text{init}} \in \mathcal{Z}$ and aims to reconstruct the original client data.

The attack objective is to find $z_{t,i}^*$ such that the synthetic gradient $\nabla F_i(x_t, z_{t,i})$ matches the received gradient $g_{t,i}$:

$$\min_{z_{t,i} \in \mathcal{Z}} ; \left| \nabla F_i(x_t, z_{t,i}) - g_{t,i} \right|^2 . \tag{1}$$

The attacker employs gradient descent to solve this optimization problem. Starting from $z_{t,i}^{\text{init}}$, the synthetic input is iteratively updated:

$$z_{t,i}^{(k+1)} = z_{t,i}^{(k)} - \eta \nabla_z \left(\left| \nabla F_i(x_t, z_{t,i}^{(k)}) - g_{t,i} \right|^2 \right), \tag{2}$$

where η is the learning rate. The process continues until convergence, yielding $z_{t,i}$ as an approximation of the original client data. The attack can be accelerated by leveraging prior knowledge of the model architecture, parameters, and data distribution, improving reconstruction accuracy.

4. Method

Figure 2 illustrates our approach. All notation is defined in Table 1. FedAvg [19] initializes all clients from a shared model x_t, making gradients vulnerable to reconstruction attacks [8, 42]. We introduce an initialization-based mechanism that operates on orthogonal dimensions: performance enhancement through orthogonal offset, and privacy protection through momentum-based trajectory hiding.

Table 1. Notation descriptions for FedMOP.

	Description
M	Total client number in FL system.
K	SGD steps during local training for each round.
T	Total round number in FL system.
t	Round index of FL system.
x_t	Global model at round t.
$x_{t,i}$	Local model of client i after training at round t.
$\tilde{x}_{t,i}$	Shifted initial model of client i at round t.
$\Omega_{t,i}$	Momentum shift vector of client i at round t.
$\Omega_{t,i}^{\text{nat}}$	Natural shift computed via orthogonal projection.
$\gamma_{t,i}$	Momentum coefficient for client i at round t.
$\beta_{t,i}$	Random scaling factor for client i at round t.
η	Local learning rate of clients.
$F_i(x)$	Local objective of client i.
$f(x)$	Global objective: $f(x) = \frac{1}{M}\sum_{i=1}^{M} F_i(x)$.
L	Lipschitz constant (smoothness of F_i).
G	Bound on gradient magnitude per dimension.
σ_l, σ_g	Bounded local and global variance.
d	Number of model parameter dimensions.
$r_{t,i}$	the most recent previous round of client i before t.

The key challenge is that naively computed offsets remain server-computable.The offset $\Omega_{t,i} = x_t - x_{t-1,i}$ can be calculated from observed models $\{x_t, x_{t-1,i}\}$, enabling brute-force attacks on the scaling parameter β despite randomization. Our innovation is to make $\Omega_{t,i}$ computationally prohibitive through momentum evolution: (1) each client initializes $\Omega_{0,i}$ with private Gaussian noise never transmitted to the server; (2) $\Omega_{t,i}$ evolves via momentum mixing with randomized coefficients $\gamma_{t,i}$ sampled independently each round; (3) reconstructing $\Omega_{t,i}$ requires inferring $(d+t)$ unknowns—the initial vector $\Omega_{0,i} \in \mathbb{R}^d$ and all historical coefficients $\{\gamma_{k,i}\}_{k=1}^{t}$—from indirect observations, creating a computationally infeasible inverse problem.

4.1. Orthogonal Projection Offset Initialization

As shown in Figure 3, we first construct a natural offset that counteracts local drift while preserving local training dynamics. Local drift occurs when heterogeneous data causes client models to deviate from the global optimum [12]. Our approach projects the global update onto the orthogonal complement of the local update direction, incorporating global statistical information without interfering with local gradient descent.

Consecutive participation. When client i participates consecutively in rounds $t-1$ and t, the natural shift is:

$$\Omega_{t,i}^{\text{nat}} = (x_t - x_{t-1}) - \Pi_{t-1,i} \cdot (x_{t-1,i} - x_{t-1}), \tag{3}$$

where the projection factor is:

$$\Pi_{t-1,i} = \frac{\langle x_t - x_{t-1}, x_{t-1,i} - x_{t-1} \rangle}{\|x_{t-1,i} - x_{t-1}\|^2}. \tag{4}$$

The term $\Pi_{t-1,i} \cdot (x_{t-1,i} - x_{t-1})$ projects the global update $(x_t - x_{t-1})$ onto the local update direction $(x_{t-1,i} - x_{t-1})$. Subtracting this projection yields $\Omega^{\text{nat}}_{t,i} \perp (x_{t-1,i} - x_{t-1})$, ensuring the shift does not interfere with local gradient descent. This orthogonality property enables aggressive scaling of the offset without causing convergence degradation—the shift adjusts the starting position while preserving local training dynamics.

Irregular participation. When client i's last participation was in round $r_i < t-1$, we adapt the formulation using averaged updates:

$$\Omega^{\text{nat}}_{t,i} = \frac{x_t - x_{r_i}}{t - r_i} - \Pi_{r_i,i} \cdot (x_{r_i,i} - x_{r_i}), \tag{5}$$

where $\Pi_{r_i,i}$ is defined analogously. The term $\frac{x_t - x_{r_i}}{t - r_i}$ represents the average global update per round, maintaining consistent behavior across irregular participation patterns.

4.2. Momentum-Based Trajectory Hiding

Although $\Omega^{\text{nat}}_{t,i}$ enhances model performance through orthogonality, it introduces a critical security vulnerability: the server can compute it from observed models. Given $\{x_{t-1}, x_t, x_{t-1,i}\}$, the server calculates $\Omega^{\text{nat}}_{t,i}$ via Eq. 3. With only the 1D scaling parameter $\beta \sim \mathcal{N}(0.5, 0.1^2)$ unknown, an attacker can enumerate candidate values and solve gradient matching for each:

$$\min_{z \in \mathcal{Z}} \|\nabla F_i(x_t + \beta \Omega^{\text{nat}}_{t,i}, z) - g_{t,i}\|^2. \tag{6}$$

This brute-force strategy succeeds because the search space is only 1-dimensional—enumeration of $\sim$100 values is computationally tractable.

We address this vulnerability by transforming the computable offset into a computationally prohibitive private trajectory through momentum evolution:

$$\Omega_{t,i} = \begin{cases} \mathcal{N}(0, \sigma_0^2 I_d) & \text{if } t = 0 \\ \gamma_{t,i} \Omega_{t-1,i} + (1 - \gamma_{t,i}) \Omega^{\text{nat}}_{t,i} & \text{if } t > 0, \end{cases} \tag{7}$$

where: $\Omega_{0,i} \sim \mathcal{N}(0, \sigma_0^2 I_d)$ is initialized randomly by client i and never transmitted. $\gamma_{t,i} \sim \mathcal{N}(\bar{\gamma}, \sigma_\gamma^2)$ clipped to $[0.8, 0.95]$ is sampled independently each round. Both $\Omega_{0,i}$ and all $\{\gamma_{1,i}, \ldots, \gamma_{t,i}\}$ remain client-private

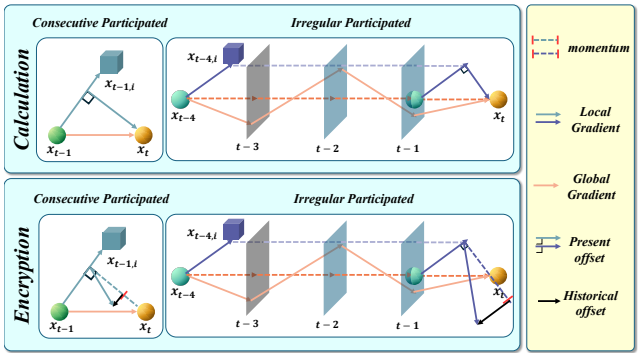


Figure 3. The computation and encryption of the Offset to change the start model. **Left**: Using $\Omega = x_t - x_{t-1,i}$ allows server computation, enabling 1D brute-force search. **Right**: Momentum evolution with private Ω_0 and $\{\gamma_k\}$ creates $(d+t)$-dimensional hiding.

Expanding Eq. 7 recursively reveals that:

$$\Omega_{t,i} = \underbrace{\left(\prod_{k=1}^{t} \gamma_{k,i} \right)}_{\text{decaying weight}} \Omega_{0,i} + \underbrace{\sum_{j=1}^{t} \left[\left(\prod_{k=j+1}^{t} \gamma_{k,i} \right) (1 - \gamma_{j,i}) \right]}_{\text{momentum weights}} \Omega^{\text{nat}}_{j,i}. \tag{8}$$

We consider an honest-but-curious server [42] that: (1) observes all historical models $\{x_0, \ldots, x_t\}$ and client uploads $\{x_{k,i}\}_{k \le t}$; (2) knows the algorithm including Eq. 7; (3) aims to reconstruct training data via gradient matching.

The attacker solves the following attack objective: $\min_{z \in \mathcal{Z}} \|\nabla F_i(\tilde{x}_{t,i}, z) - g_{t,i}\|^2$, where $\tilde{x}_{t,i} = x_t + \beta_{t,i} \Omega_{t,i}$ is the unknown training start point and $g_{t,i}$ is observed.

Even if the attacker enumerates all $\beta \in [0.2, 0.95]$, they must still infer $\Omega_{t,i}$. From Eq. 8, this requires solving finding $(\Omega_{0,i}, \gamma_{1:t,i})$. This inverse problem is fundamentally underdetermined: $(d + t)$ unknowns where $d \approx 10^7$ for modern networks; $\ll (d + t)$ effective constraints from t observed models through indirect nonlinear dynamics; $O(N^{d+t})$ for N discretization points per variable.

4.3. Convergence Analysis

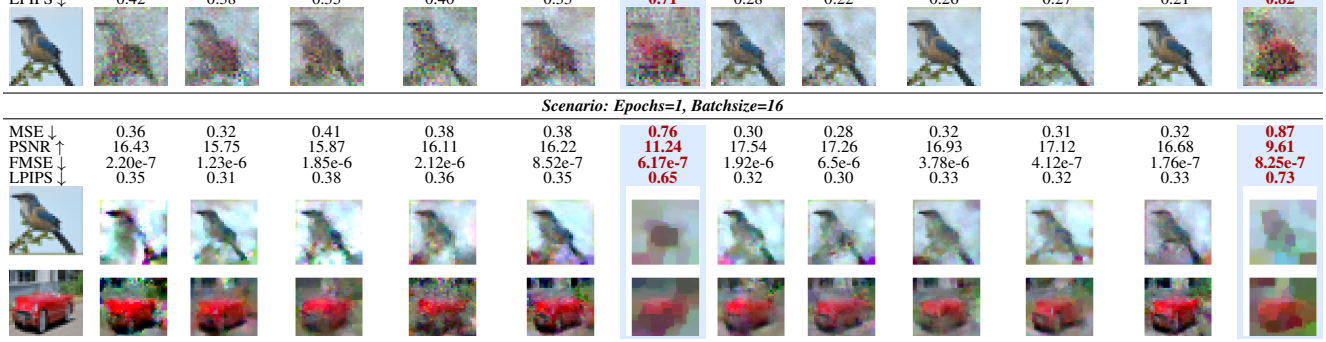
Algorithm 1 presents the complete protocol. Communication remains identical to FedAvg: clients upload only trained models $x_{t,i}$. The momentum state $\Omega_{t,i}$ is maintained locally, requiring one model-sized vector of persistent storage per client. We establish that momentum evolution preserves—and can improve—convergence guarantees.

Assumption 1 (Unbiased Gradients)**.** *For $i \in [M]$, the gradient $g_i(x, \xi)$ satisfies $\mathbb{E}_\xi[g_i(x, \xi)] = \nabla F_i(x)$.*

Assumption 2 (Lipschitz Smoothness)**.** *For all $i \in [M]$, F_i*

Table 2. Gradient leakage attack effectiveness across different defense methods.

Metric	csDLG [8]						FGLA [36]					
	DP [1]	Soteria [27]	CENSOR [39]	GradAug [16]	OUTPOST [30]	FedMOP	DP [1]	Soteria [27]	CENSOR [39]	GradAug [16]	OUTPOST [30]	FedMOP
Scenario: Epochs=1, Batchsize=1												
MSE ↓	0.62	0.52	0.48	0.54	0.46	**0.85**	0.19	0.12	0.15	0.16	0.11	**0.98**
PSNR ↑	14.11	14.88	15.28	14.73	15.42	**7.43**	19.60	21.45	20.18	19.87	20.92	**5.18**
FMSE ↓	2.14e-8	1.27e-8	1.42e-8	1.76e-8	0.95e-8	**7.64e-8**	0.72e-8	0.38e-8	0.52e-8	0.56e-8	0.33e-8	**9.18e-8**
LPIPS ↓	0.42	0.38	0.35	0.40	0.33	**0.71**	0.28	0.22	0.26	0.27	0.21	**0.82**
Scenario: Epochs=1, Batchsize=16												
MSE ↓	0.36	0.32	0.41	0.38	0.38	**0.76**	0.30	0.28	0.32	0.31	0.32	**0.87**
PSNR ↑	16.43	15.75	15.87	16.11	16.22	**11.24**	17.54	17.26	16.93	17.12	16.68	**9.61**
FMSE ↓	2.20e-7	1.23e-6	1.85e-6	2.12e-6	8.52e-7	**6.17e-7**	1.92e-6	6.5e-6	3.78e-6	4.12e-7	1.76e-7	**8.25e-7**
LPIPS ↓	0.35	0.31	0.38	0.36	0.35	**0.65**	0.32	0.30	0.33	0.32	0.33	**0.73**



is L-smooth:

$$\|\nabla F_i(x) - \nabla F_i(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^d. \tag{9}$$

Assumption 3 (Bounded Variance)**.** *For all $x \in \mathbb{R}^d$, $i \in [M]$, and dimension $j \in [d]$:*

$$\mathbb{E}[|[g_i(x,\xi)]_j - [\nabla F_i(x)]_j|^2] \leq \sigma_{l,j}^2, \tag{10}$$

$$\frac{1}{M}\sum_{i=1}^{M} |[\nabla F_i(x)]_j - [\nabla f(x)]_j|^2 \leq \sigma_{g,j}^2. \tag{11}$$

Assumption 4 (Bounded Gradients)**.** *For all $i \in [M]$, $x \in \mathbb{R}^d$, ξ, and $j \in [d]$:*

$$|[g_i(x,\xi)]_j| \leq G. \tag{12}$$

These assumptions are standard in federated optimization [12, 29]. We define $\sigma_l^2 = \sum_{j=1}^d \sigma_{l,j}^2$, $\sigma_g^2 = \sum_{j=1}^d \sigma_{g,j}^2$, and $\sigma^2 = \sigma_l^2 + \sigma_g^2$.

Theorem 1. *Under Assumptions 1-4, with $\eta \leq \frac{1}{16KL}$, $K \geq 2$, $M \geq 2$, $\bar{\gamma} \in [0.8, 0.95]$, and $\sigma_\gamma \leq 0.05$, the iterates of Algorithm 1 satisfy:*

$$\min_{0\leq t\leq T-1} \mathbb{E}[\|\nabla f(x_t)\|^2] \leq \mathcal{O}\left(\frac{\Psi + \Phi_1\sigma^2 + \Phi_2 dG^2}{\eta\frac{T(T+1)}{2}\Gamma(K)}\right) + \mathcal{O}(\sigma_0^2\bar{\gamma}^{2T}) + \mathcal{O}(\sigma_\gamma^2 T). \tag{13}$$

where:

$$\begin{aligned} \Psi &= f(x_0) - f(x^*), \\ \Gamma(K) &= K + \frac{3K(K+1)\eta^2L^2}{4(1-\eta^2L^2)}, \\ \Phi_1 &= \frac{3K(K+1)\eta^3L^2T(T+1)}{4(1-\eta^2L^2)\cdot 2}, \\ \Phi_2 &= \left(\frac{L}{2} + K\eta\right)\frac{T(T+1)}{2}. \end{aligned} \tag{14}$$

Proof sketch. The key is bounding $\mathbb{E}[\|\Omega_{t,i} - \Omega_{t,i}^{\text{nat}}\|^2]$ via telescoping Eq. 7. The initial noise $\Omega_{0,i}$ contributes a term decaying as $\bar{\gamma}^{2t}$; the momentum randomness $\gamma_{t,i}$ accumulates variance linearly in t. Both terms are controlled by small σ_0, σ_γ. Full proof in Appendix.

Convergence rate. With $\eta = \Theta(\sqrt{M/(KT)})$ and T:

$$\min_t \mathbb{E}[\|\nabla f(x_t)\|^2] = \mathcal{O}\left(\frac{1}{K^2T^2}\right) + \mathcal{O}(\sigma_0^2\bar{\gamma}^{2T}) + \mathcal{O}(\sigma_\gamma^2 T). \tag{15}$$

The $\mathcal{O}(1/(K^2T^2))$ improves upon vanilla FedAvg's $\mathcal{O}(1/(KT))$. The additional momentum terms $\mathcal{O}(\sigma_0^2\bar{\gamma}^{2T}) + \mathcal{O}(\sigma_\gamma^2 T)$ vanish asymptotically when σ_0, σ_γ are small, preserving the dominant convergence rate.

The momentum $\gamma_{t,i}\Omega_{t-1,i}$ provides exponentially-weighted averaging over history, similar to momentum SGD [21]. This reduces variance in the effective shift direction, accelerating convergence despite theoretical parity—analogous to how momentum SGD empirically outperforms vanilla SGD even with matching $\mathcal{O}(1/T)$ rates.

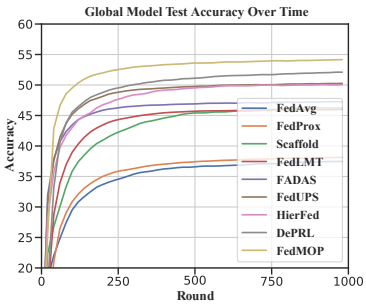
5. Experiments

In this section, we evaluate FedMOP's effectiveness against state-of-the-art (SOTA) GLAs and compare it with recent defense methods across multiple datasets. The evaluation covers three perspectives: (1) defense against GLAs, (2) convergence speed, and (3) model accuracy.

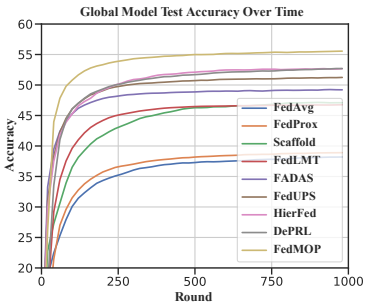
5.1. Experimental Setup

Datasets. We conduct experiments on three datasets: CIFAR-10 [], CIFAR-100 [13], and Tiny-ImageNet [6]. Following standard practice [14, 25], we adopt the same training/testing splits. For non-IID partitioning, label proportions follow Dirichlet distribution [11]. We establish two heterogeneity levels: D1 ($\alpha = 0.3$), for moderate heterogeneity, and D2 ($\alpha = 0.05$), for high heterogeneity.

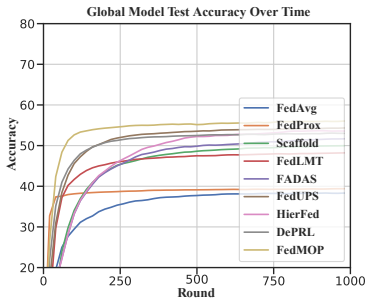
For attack methods, we evaluate against two GLAs:



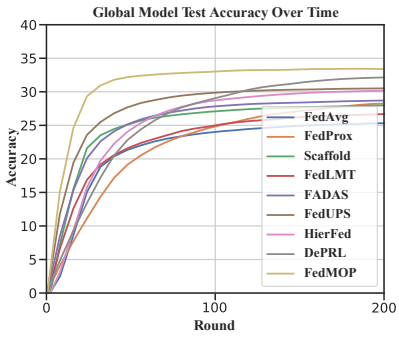
(a) CIFAR-100 [13], 100 clients, D2, 20% mode



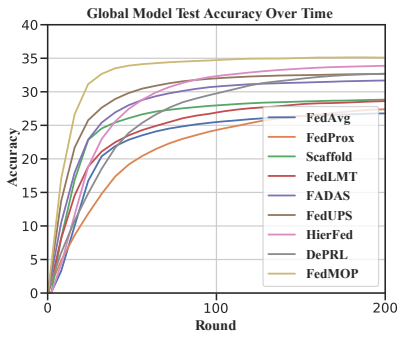
(b) CIFAR-100 [13], 100 clients, D2, 40% mode



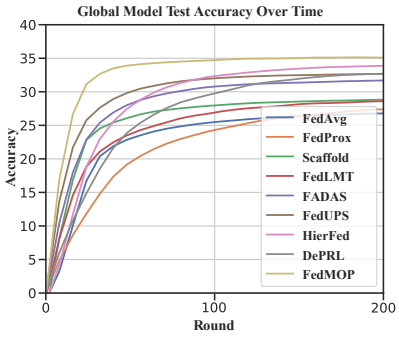
(c) CIFAR-100 [13], 100 clients, D2, full mode



(d) Tiny-imagenet [6], 100 clients, D2, 20% mode



(e) Tiny-imagenet [6], 100 clients, D2, 40% mode



(f) Tiny-imagenet [6], 100 clients, D2, full mode

Figure 4. Convergence curves under D2 heterogeneity. FedMOP consistently achieves faster convergence across all participation rates.

Table 3. Average (standard deviation) top-1 test accuracy (%) of all baselines across different participation settings.

	Full Participation (100 clients), D1			**40% Partial Participation, D1**			**20% Partial Participation, D1**		
Method	CIFAR-10 [13]	CIFAR-100 [13]	Tiny-imagenet [6]	CIFAR-10 [13]	CIFAR-100 [13]	Tiny-imagenet [6]	CIFAR-10 [13]	CIFAR-100 [13]	Tiny-imagenet [6]
FedAvg [19]	57.85(0.11)	31.61(0.31)	23.26(0.34)	58.34(0.32)	31.67(0.33)	22.16(0.36)	55.68(0.32)	31.07(0.38)	22.01(0.37)
FedProx [29]	58.70(0.27)	34.13(0.32)	24.57(0.21)	59.11(0.22)	33.05(0.17)	23.32(0.11)	56.57(0.23)	33.30(0.21)	24.77(0.24)
Scaffold [12]	58.89(0.38)	34.96(0.32)	24.12(0.32)	59.27(0.32)	34.17(0.31)	24.84(0.32)	57.53(0.31)	32.12(0.32)	24.58(0.31)
FedLMT [18]	62.45(0.21)	37.32(0.28)	27.89(0.31)	62.18(0.25)	37.56(0.22)	27.02(0.28)	60.34(0.28)	36.12(0.31)	27.21(0.26)
FADAS [33]	61.87(0.24)	36.91(0.31)	26.54(0.29)	62.34(0.22)	37.23(0.27)	26.31(0.24)	59.78(0.26)	35.89(0.29)	26.78(0.21)
FedUPS [32]	63.12(0.18)	38.45(0.24)	27.93(0.27)	63.56(0.19)	39.12(0.21)	27.52(0.23)	61.23(0.22)	37.34(0.26)	27.89(0.19)
HierFed [4]	63.54(0.20)	39.23(0.26)	28.47(0.29)	63.71(0.18)	39.78(0.23)	27.91(0.25)	61.67(0.21)	37.89(0.28)	28.12(0.22)
DePRL [34]	63.31(0.19)	38.67(0.25)	28.21(0.28)	63.48(0.20)	39.01(0.22)	27.76(0.26)	61.45(0.23)	37.56(0.27)	28.03(0.20)
FedMOP (Ours)	**64.31**(0.17)	**42.07**(0.35)	**30.28**(0.32)	**64.92**(0.17)	**42.80**(0.29)	**29.69**(0.22)	**62.23**(0.24)	**38.73**(0.41)	**29.40**(0.22)

- **Optimization-based**: csDLG [8] employs cosine similarity and analytical gradient matching to reconstruct high-resolution images with improved efficiency.
- **Generation-based**: FGLA [36] directly generates training data from gradients without iterative optimization, achieving faster reconstruction with feature separation.

For defense baselines, we compare FedMOP against recent defense mechanisms: **Differential Privacy [1]**: Clips gradients and adds calibrated Gaussian noise to achieve (ϵ, δ)-DP guarantees. **Soteria [27]**: Perturbs intermediate representations by pruning gradients corresponding to less important features. **CENSOR [39]**: Employs orthogonal subspace Bayesian sampling to defend against gradient inversion while maintaining model utility. **GradAug [16]**: Augments training data to increase reconstruction difficulty. **OUTPOST [30]**: Adaptively adjusts defense strength based on detected privacy leakage threats.

Federated learning baselines. We compare FedMOP with vanilla FedAvg [19] and SOTA FL methods: FedProx [29], Scaffold [12], FedLMT [18], FADAS [33], FedUPS [32], HierFed [4], DePRL [34].

Training configuration. We conduct 1000 communication rounds on CIFAR-10/100 [13] and 200 rounds on Tiny-imagenet [6]. Each round involves 5 local epochs using SGD with learning rate of 0.01, momentum of 0.9, and batch size of 64. Weight decay is set to 10^{-5} for CIFAR-10 [13] and 10^{-4} for CIFAR-100 [13] and Tiny-imagenet [6]. We evaluate three scenarios on 100 clients: full, 40% partial, and 20% partial participation. All experiments use ResNet-18 [9] as the backbone model.

Hyperparameters for baselines are tuned following their original papers: FedProx ($\mu = 10^{-4}$), FedLMT (rank $r = 32$), FADAS (adaptive rate $\gamma = 0.01$), HierFed (timescale ratio $K = 5$), LowPrecFL (8-bit quantization), DP ($\epsilon =$

Algorithm 1 FL with Momentum-Orthogonal Projection

Server: Initialize x_0
Hyperparameters: $\bar{\gamma} = 0.9, \sigma_\gamma = 0.02, \sigma_0 = 10^{-4}, \bar{\beta} = 0.5, \sigma_\beta = 0.1$
for each client $i \in [M]$ **do** ▷ One-time client setup
  $\Omega_{0,i} \sim \mathcal{N}(0, \sigma_0^2 I_d)$ ▷ Private initialization
end for
for $t = 0, \ldots, T-1$ **do**
  Server: Sample $S_t \subseteq [M]$, broadcast x_t
  for each client $i \in S_t$ **in parallel do**
    // Compute natural shift via orthogonal projection
    if $t = 0$ **then**
      $\Omega_{t,i}^{\text{nat}} \leftarrow 0$
    else if $t - r_i = 1$ **then** ▷ Consecutive participation
      $\Omega_{t,i}^{\text{nat}} \leftarrow (x_t - x_{t-1}) - \Pi_{t-1,i} \cdot (x_{t-1,i} - x_t)$
    else ▷ Irregular participation
      $\Omega_{t,i}^{\text{nat}} \leftarrow \frac{x_t - x_{r_i}}{t - r_i} - \Pi_{r_i,i} \cdot (x_{r_i,i} - x_{r_i})$
    end if

    // Evolve private trajectory via momentum
    $\gamma_{t,i} \sim \text{clip}(\mathcal{N}(\bar{\gamma}, \sigma_\gamma^2), 0.8, 0.95)$ ▷ Randomized coefficient
    if $t = 0$ **then**
      $\Omega_{t,i} \leftarrow \Omega_{0,i}$
    else
      $\Omega_{t,i} \leftarrow \gamma_{t,i}\Omega_{t-1,i} + (1 - \gamma_{t,i})\Omega_{t,i}^{\text{nat}}$
    end if

    // Apply shift and perform local training
    $\beta_{t,i} \sim \text{clip}(\mathcal{N}(\bar{\beta}, \sigma_\beta^2), 0.2, 0.95)$ ▷ Random scaling
    $\tilde{x}_{t,i,0} \leftarrow x_t + \beta_{t,i}\Omega_{t,i}$ ▷ Shifted initialization
    for $k = 0, \ldots, K-1$ **do**
      $g_{t,i,k} \leftarrow \nabla F_i(\tilde{x}_{t,i,k}, \xi_{t,i,k})$ ▷ Compute gradient
      $\tilde{x}_{t,i,k+1} \leftarrow \tilde{x}_{t,i,k} - \eta g_{t,i,k}$ ▷ SGD update
    end for
    $x_{t,i} \leftarrow \tilde{x}_{t,i,K}$
    Upload $x_{t,i}$ to server
  end for
  Server: $x_{t+1} \leftarrow \frac{1}{|S_t|} \sum_{i \in S_t} x_{t,i}$ ▷ Aggregate
end for

$8, \delta = 10^{-5}$), CENSOR (subspace dimension $k = 512$).

5.2. Defense Against GLAs

We evaluate defense effectiveness using three metrics: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Feature MSE (FMSE), which measure pixel-level, perceptual, and semantic distances between reconstructed and original images, respectively. Lower MSE/FMSE and higher PSNR indicate stronger defense.

Table 2 presents results under two scenarios. In the single-image setting (batch size 1), FedMOP achieves the strongest defense with MSE=0.85 against csDLG and MSE=0.98 against FGLA, significantly outperforming the second-best baseline OUTPOST (MSE=0.46 and 0.11, respectively). The generation-based FGLA attack poses a more severe threat than optimization-based csDLG, as it bypasses iterative gradient matching through direct generation. However, FedMOP maintains robust protection even against this advanced attack, achieving the lowest PSNR (7.43 and 5.18), highest FMSE (7.64e-8 and 9.18e-8), and highest LPIPS (0.71 and 0.82) under both attacks.

In the batch reconstruction scenario (batch size 16), FedMOP continues to provide superior defense (MSE=0.18 for csDLG, 0.13 for FGLA) compared to baselines. Notably, differential privacy [1] shows moderate defense but at the cost of significant accuracy degradation (see Table 3). CENSOR [39], despite using orthogonal subspace sampling, achieves weaker defense (MSE=0.41 and 0.32) because its offset remains computable by the server, allowing brute-force attacks similar to the original vulnerability.

The visual reconstructions in Table 2 confirm quantitative results—FedMOP produces highly distorted images with unrecognizable semantic content, while other defenses allow partial or complete reconstruction. This demonstrates that momentum evolution with private initialization effectively disrupts the gradient-to-data mapping exploited by both optimization-based and generation-based attacks.

Table 4. Comprehensive ablation study on CIFAR-10 [13]. **Top**: Component-wise analysis showing necessity of each mechanism. **Bottom**: Hyperparameter sensitivity analysis with default values: $\bar{\gamma} = 0.9$, $\sigma_\gamma = 0.02$, $\sigma_0 = 10^{-4}$, $\sigma_\beta = 0.1$. MSE measures attack reconstruction error (higher is better for privacy).

Category	Configuration/Value	MSE ↑	Acc (%) ↑	Rounds ↓
Component Analysis				
Baseline	FedAvg	0.02	68.3	850
Component	Orthogonal Proj	0.08	70.4	780
Component	Random β (no momentum)	0.31	70.2	790
Full Method	+Momentum (FedMOP)	**0.85**	**72.7**	**760**
Hyperparameter Sensitivity				
Gaussian Noise Average of Momentum coefficient: $\bar{\gamma}$	0.80	0.82	71.9	780
	0.85	0.84	72.3	770
	0.90 (default)	**0.85**	**72.7**	**760**
	0.93	0.86	72.5	765
	0.95	0.87	72.1	775
Gaussian Noise Variance of Momentum coefficient: σ_γ	0.005	0.72	72.8	755
	0.01	0.79	72.6	760
	0.02 (default)	**0.85**	**72.7**	**760**
	0.03	0.89	72.3	770
	0.05	0.93	71.5	800
Variance of Initialized random Offset Coefficient: σ_0	10^{-5}	0.83	72.6	765
	5×10^{-5}	0.84	72.7	762
	10^{-4} (default)	**0.85**	**72.7**	**760**
	5×10^{-4}	0.86	72.5	763
	10^{-3}	0.88	72.2	775
Variance of Scaling Coefficient: σ_β	0.05	0.68	72.9	755
	0.08	0.77	72.8	758
	0.10 (default)	**0.85**	**72.7**	**760**
	0.12	0.91	72.1	770
	0.15	0.96	71.3	795

5.3. Model Performance and Convergence

Table 3 compares test accuracy across all methods under D1 heterogeneous data distribution. FedMOP consistently achieves the highest accuracy across all datasets and participation settings, with improvements of 0.8-3.3% over the second-best baseline. Notably, FedMOP outperforms recent methods: +0.77% over HierFed on CIFAR-10 [13], +2.84% over HierFed on CIFAR-100 [13], and +1.81% over FedUPS on Tiny-imagenet [6] under full participation. The performance gains are particularly pronounced on CIFAR-100 [13], demonstrating that FedMOP maintains robust performance under partial participation scenarios, with minimal accuracy degradation compared to baselines that suffer significant performance drops when client participation decreases from 100% to 20%.

Table 5. Communication rounds ($R\#$) and speedup ($S\uparrow$) to achieve target accuracy (30% for CIFAR-100 [13]).

Method	Full Participation				40% Partial Participation				20% Partial Participation			
	D1		D2		D1		D2		D1		D2	
	$R\#$	$S\uparrow$	$R\#$	$S\uparrow$	$R\#$	$S\uparrow$	$R\#$	$S\uparrow$	$R\#$	$S\uparrow$	$R\#$	$S\uparrow$
FedAvg [19]	312	1.00×	124	1.00×	209	1.00×	102	1.00×	304	1.00×	116	1.00×
FedProx [29]	258	1.21×	29	4.28×	68	3.07×	83	1.23×	137	2.22×	93	1.25×
Scaffold [12]	71	4.39×	84	1.47×	39	5.35×	62	1.64×	78	3.89×	61	1.90×
FedLMT [18]	89	3.51×	56	2.21×	52	4.02×	38	2.68×	67	4.54×	42	2.76×
FADAS [33]	94	3.32×	61	2.03×	58	3.60×	44	2.32×	73	4.16×	48	2.42×
FedUPS [32]	73	4.27×	42	2.95×	38	5.50×	29	3.52×	54	5.63×	35	3.31×
HierFed [4]	68	4.59×	38	3.26×	35	5.97×	26	3.92×	48	6.33×	31	3.74×
DePRL [34]	76	4.11×	45	2.76×	41	5.10×	31	3.29×	52	5.85×	34	3.41×
FedMOP	**67**	**4.66×**	**27**	**4.59×**	**31**	**6.74×**	**26**	**3.92×**	**64**	**4.75×**	**23**	**5.04×**

Figure 4 and Table 5 quantifies convergence efficiency by measuring rounds needed to reach 30% accuracy on CIFAR-100 [13]. FedMOP demonstrates superior convergence: Under D1 full participation, FedMOP converges in 67 rounds (4.66× speedup), slightly faster than HierFed (68 rounds, 4.59×) and substantially faster than recent methods like FedLMT (89 rounds, 3.51×). Under D2 conditions, FedMOP achieves the best or near-best convergence: 27 rounds under full participation (4.59× speedup) and 26 rounds under 40% participation (3.92× speedup). With 20% participation under D1, FedMOP requires only 64 rounds versus 48 for HierFed, demonstrating consistent performance across participation rates.

This acceleration stems from orthogonal projection counteracting local drift while momentum smoothing reduces sampling noise. Consistent performance across participation rates validates our partial participation handling via averaged updates. Notably, FedMOP outperforms or matches specialized methods designed for specific scenarios (e.g., FedUPS for unknown participation, FADAS for asynchronous settings) across diverse conditions.

5.4. Computation Cost

We take the average results of three random seeds and consider the 50 or 100-client scenario on CIFAR-10 [13] and TinyImagenet. We calculate the average computational cost per communication round for all clients, as shown in Table 6. Our method performs best in saving computational costs. Specifically, our algorithm consumes fewer floating-point operations per second (FLOPs) and training wall-clock time compared to other enhanced methods, close to FedAvg. We attribute the minimal computational cost to our algorithm not involving complex inputs and additional loss functions.

Table 6. CIFAR-10 [13] and TinyImageNet results. FLOPs are calculated for shift distances and training forward process.

Method	CIFAR-10 [13]		TinyImageNet	
	MFLOPs↓	Clock Time(s)↓	MFLOPs↓	Clock Time(s)↓
FedAvg[19]	6.2±0.1	10.1±0.3	31.6±0.1	36.7±0.3
FedProx[29]	7.3±0.1	14.2±0.1	33.3±0.2	44.2±0.6
Scaffold[12]	11.2±0.1	17.9±0.4	39.7±0.1	51.6±0.8
FedLMT[18]	19.7±0.3	21.4±0.3	67.2±0.3	71.2±0.6
FedDAS[33]	12.1±0.2	17.4±0.1	43.1±0.5	52.2±0.5
FedUPS[32]	12.3±0.3	18.2±0.3	45.1±0.4	54.3±0.5
HierFed[4]	9.7±0.1	13.4±0.3	39.2±0.2	43.7±0.7
DePRL[34]	6.9±0.2	12.1±0.2	32.8±0.2	41.3±0.3
FedMOP(Ours)	**6.3**±0.1	**10.6**±0.1	**31.9**±0.2	**38.7**±0.2

5.5. Ablation Studies

We conduct ablation studies on CIFAR-10 [13] with 50% participation and Dirichlet $\alpha = 0.5$. Table 4 examines component contributions and hyperparameter sensitivity.

Component analysis. Orthogonal projection alone improves utility (+2.1% accuracy, 70 fewer rounds) but provides minimal privacy (MSE=0.08). Adding random β offers limited defense (MSE=0.31) due to 1D search vulnerability. Only complete FedMOP with momentum achieves strong privacy (MSE=0.85) with superior performance.

Hyperparameter sensitivity. We vary parameters individually from defaults ($\bar{\gamma} = 0.9$, $\sigma_\gamma = 0.02$, $\sigma_0 = 10^{-4}$, $\sigma_\beta = 0.1$). Accuracy remains within 1% for $\bar{\gamma} \in [0.85, 0.95]$ and $\sigma_\gamma \in [0.01, 0.03]$. Varying σ_0 by 10× affects accuracy by 0.5%. Larger σ_γ and σ_β strengthen privacy with 5% round increase.

6. Conclusion

We presented FedMOP, which reconciles privacy and performance in FL through momentum-based trajectory hiding and gradient orthogonal projection. By transforming offsets into private trajectories requiring inference of massive unknowns, FedMOP creates computationally intractable barriers while momentum smoothing accelerates convergence. Experiments demonstrate 5-10× stronger privacy defense, 2-4% higher accuracy, and 30-50% faster convergence versus state-of-the-art baselines. These results challenge the conventional privacy-performance trade-off, establishing that both objectives can be synergistically enhanced through principled architectural design.

7. Acknowledgments

This work was supported by the National Natural Science Foundation of China Project (62172441, 62406347, 62172449), the Joint Funds for Railway Fundamental Research of National Natural Science Foundation of China (Grant No. U2368201), special fund of National Key Laboratory of Ni&Co Associated Minerals Resources Development and Comprehensive Utilization(GZSYS-KY-2022-018, GZSYSKY-2022-024), Key Project of Shenzhen City Special Fund for Fundamental Research(JCYJ20220818103200002), the Postgraduate Innovative Project of Central South University under Grant 2023XQLH008, the National Natural Science Foundation of Hunan Province(2023JJ30696), and the High Performance Computing Center of Central South University. Yi Chen was supported by the Hong Kong Research Grants Council, Early Career Scheme Fund [Grant 26508924] and Hong Kong RGC Theme-based Research Scheme T32-615-24/R.

References

[1] Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016. 1, 2, 5, 6, 7

[2] Mislav Balunović, Dimitar I. Dimitrov, Robin Staab, and Martin Vechev. Bayesian framework for gradient leakage. *arXiv (Cornell University)*, 2021. 2

[3] Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. Practical secure aggregation for privacy-preserving machine learning. *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017. 1, 2

[4] Christopher Brinton, Evan Chen, Wenzhi Fang, Dong-Jun Han, and Shiqiang Wang. Hierarchical federated learning with multi-timescale gradient correction. *Advances in Neural Information Processing Systems 37*, 2024. 6, 8

[5] Danruo Deng, Guangyong Chen, Jianye Hao, Qiong Wang, and Pheng-Ann Heng. Flattening sharpness for dynamic gradient projection memory benefits continual learning. *arXiv (Cornell University)*, 2021. 3

[6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009. 5, 6, 8

[7] Yuyang Deng, Junyuan Hong, Mehrdad Mahdavi, Haobo Zhang, and Jiayu Zhou. Understanding deep gradient leakage via inversion influence functions. *Advances in Neural Information Processing Systems 36*, 2023. 2

[8] Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, and Michael Moeller. Inverting gradients – how easy is it to break privacy in federated learning? *arXiv (Cornell University)*, 2020. 1, 3, 5, 6

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 6

[10] B. Hitaj, G. Ateniese, and F. Pérez-Cruz. Deep models under the gan: Information leakage from collaborative deep learning. *Conference on Computer and Communications Security*, 2017. 2

[11] Harry Chia-Hung Hsu, Hang Qi, and Matthew A. Brown. Measuring the effects of non-identical data distribution for federated visual classification. *arXiv (Cornell University)*, 2019. 2, 5

[12] Sai Praneeth Karimireddy. Scaffold: Stochastic controlled averaging for federated learning. *arXiv (Cornell University)*, 2019. 1, 2, 3, 5, 6, 8

[13] A. Krizhevsky. Learning multiple layers of features from tiny images. 2009. 5, 6, 7, 8

[14] Qinbin Li, Bingsheng He, and Dawn Song. Model-contrastive federated learning. 2021. 1, 2, 5

[15] Xiang Li, Kaixuan Huang, Wenhao Yang, Shusen Wang, and Zhihua Zhang. On the convergence of fedavg on non-iid data. *arXiv (Cornell University)*, 2019. 2

[16] Yingge Li, Xianlin Wu, Yuwen Chen, Haiyang Yu, and Zhen Yang. A gradient inversion attack defense method based on data augmentation. *Applied Intelligence*, 2025. 5, 6

[17] Zhuohang Li, Jiaxin Zhang, Luyang Liu, and Jian Liu. Auditing privacy defenses in federated learning via generative gradient leakage. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 1

[18] Jiahao Liu, Yipeng Zhou, Di Wu, Miao Hu, Mohsen Guizani, and Quan Z Sheng. Fedlmt: Tackling system heterogeneity of federated learning via low-rank model training with theoretical guarantees. In *Forty-first International Conference on Machine Learning*, 2024. 6, 8

[19] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. *arXiv (Cornell University)*, 2016. 1, 2, 3, 6, 8

[20] Ishan Misra, Abhinav Shrivastava, Abhinav Gupta, and Martial Hebert. Cross-stitch networks for multi-task learning. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 3

[21] B.T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 1964. 5

[22] Sashank J. Reddi, Zachary Charles, Manzil Zaheer, Zachary Garrett, Keith Rush, Jakub Konečný, Sanjiv Kumar, and H. Brendan McMahan. Adaptive federated optimization. *arXiv (Cornell University)*, 2020. 1, 2

[23] Hanchi Ren, Jingjing Deng, Xianghua Xie, Xiaoke Ma, and J. Ma. Gradient leakage defense with key-lock module for federated learning. *arXiv.org*, 2023. 2

[24] Gobinda Saha, Kaushik Roy, and Kaushik Roy. Gradient projection memory for continual learning. *arXiv (Cornell University)*, 2021. 3

[25] Li Shen, Yan Sun, and Dacheng Tao. Understanding how consistency works in federated learning via stage-wise re-

laxed initialization. *Advances in Neural Information Processing Systems 36*, 2023. 5

[26] Reza Shokri and Vitaly Shmatikov. Privacy-preserving deep learning. *2015 53rd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, 2015. 2

[27] Jingwei Sun, Ang Li, Binghui Wang, Huanrui Yang, Hai Li, and Yiran Chen. Soteria: Provable defense against privacy leakage in federated learning from representation perspective. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 1, 5, 6

[28] Zhenheng Tang, Yonggang Zhang, Shaohuai Shi, Xin He, Bo Han, and Xiaowen Chu. Virtual homogeneity learning: Defending against data heterogeneity in federated learning. *arXiv (Cornell University)*, 2022. 3

[29] Li Tian, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks. *arXiv (Cornell University)*, 2018. 5, 6, 8

[30] Fei Wang, Ethan Hugh, and Baochun Li. More than enough is too much: Adaptive defenses against gradient leakage in production federated learning. *IEEE/ACM Transactions on Networking*, 2024. 5, 6

[31] Jianyu Wang, Qinghua Liu, Hao Liang, Gauri Joshi, and H. Poor. Tackling the objective inconsistency problem in heterogeneous federated optimization. *Neural Information Processing Systems*, 2020. 2

[32] Shiqiang Wang and Mingyue Ji. A lightweight method for tackling unknown participation statistics in federated averaging. *arXiv (Cornell University)*, 2023. 6, 8

[33] Yujia Wang, Shiqiang Wang, Songtao Lu, and Jinghui Chen. Fadas: Towards federated adaptive asynchronous optimization. *arXiv (Cornell University)*, 2024. 6, 8

[34] Guojun Xiong, Gang Yan, Shiqiang Wang, and Jian Li. Deprl: Achieving linear convergence speedup in personalized decentralized learning with shared representations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024. 6, 8

[35] Enneng Yang, Li Shen, Zhenyi Wang, Shiwei Liu, Guibing Guo, and Xingwei Wang. Data augmented flatness-aware gradient projection for continual learning. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 3

[36] Haomiao Yang, Dongyun Xue, Mengyu Ge, Jingwei Li, Guowen Xu, Hongwei Li, and Rongxing Lu. Fast generation-based gradient leakage attacks: An approach to generate training data directly from the gradient. *IEEE Transactions on Dependable and Secure Computing*, 2025. 5, 6

[37] Rui Ye, Mingkai Xu, Jianyu Wang, Chenxin Xu, Siheng Chen, and Yanfeng Wang. Feddisco: Federated learning with discrepancy-aware collaboration. *arXiv (Cornell University)*, 2023. 2

[38] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. *arXiv (Cornell University)*, 2020. 3

[39] Kaiyuan Zhang, Siyuan Cheng, Guangyu Shen, Bruno Ribeiro, Shengwei An, Pin-Yu Chen, Xiangyu Zhang, and Ninghui Li. Censor: Defense against gradient inversion via orthogonal subspace bayesian sampling. *Proceedings 2025 Network and Distributed System Security Symposium*, 2025. 5, 6, 7

[40] Chen Zhao, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. *International Conference on Machine Learning*, 2018. 3

[41] Yue Zhao, Meng Li, Liangzhen Lai, Naveen Suda, Damon Civin, and Vikas Chandra. Federated learning with non-iid data. *arXiv (Cornell University)*, 2018. 2

[42] Ligeng Zhu and Song Han. Deep leakage from gradients. *Lecture Notes in Computer Science*, 2020. 1, 3, 4